Course Outline

Sem		Course Name	Th	Tut	Pra	Hrs	Cr
Sem I	HSC	Humanities – 1	2	0	0	2	2
	BSC	Mathematics – 1	3	1	0	4	4
	BSC	Physics	3	0	0	3	3
	BSC	Physics – Laboratory	0	0	2	2	1
	BSC	Environment Science	2	0	0	2	2
	ESC	Workshop Practices	0	0	2	2	1
	BSC	Biology for Engineers	2	0	0	2	2
	ESC	Computer Programming -I	1	0	0	1	1
	ESC	Computer Programming -I Laboratory	0	0	2	2	1
	HSC	Universal Human Values	1	0	0	1	1
	HSC	Indian Knowledge System	2	0	0	2	2
			16	**1**	**5**	**23**	**20**
Sem II	HSC	Humanities – 2	2	0	0	2	2
	BSC	Mathematics – 2	3	1	0	4	4
	BSC	Chemistry	3	0	0	3	3
	BSC	Chemistry – Laboratory	0	0	2	2	1
	ESC	Introduction to Electrical Engineering	3	0	0	3	3
	ESC	Introduction to Electrical Engineering-Laboratory	0	0	2	2	1
	ESC	Engineering Graphics	0	0	4	4	2
	HSC	Yoga, Health & Hygiene OR NCC/NSS	0	0	2	2	1
	HSC	Organizational Behaviour	1	0	0	1	1
	ESC	Computer Programming -II	1	0	0	1	1
	ESC	Computer Programming -II Laboratory	0	0	2	2	1
			13	**1**	**12**	**26**	**20**
Sem III	Pro	Civic and Social Service Internship	0	0	0	0	1
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	PC	Digital Circuits	3	0	0	3	3
	PC	Digital Circuits Laboratory	0	0	2	2	1
	PC	Electronics Devices and Circuits	3	0	0	3	3
	PC	Electronics Devices and Circuits Laboratory	0	0	2	2	1
	PC	Networks and Systems	3	1	0	4	4
	BSC	Math-3	3	1	0	4	4
			15	**2**	**4**	**21**	**20**
Sem IV	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 – Laboratory (Department Specific)	0	0	2	2	1
	OE	Open Elective 1 (From Other School)	3	0	0	3	3
	PC	Analog Electronics	3	0	0	3	3
	PC	Analog Electronics Laboratory	0	0	2	2	1
	PC	Analog Communication	3	0	0	3	3
	PC	Analog Communication Laboratory	0	0	2	2	1
	PC	Electromagnetics and Transmission Lines	3	1	0	4	4
	PC	Digital Signal Processing	3	0	0	3	3
	PC	Digital Signal Processing Laboratory	0	0	2	2	1
			17	**1**	**8**	**26**	**22**
Sem V	OE	Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3

	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Control Systems	3	0	0	3	3
	PC	Control Systems Lab	0	0	2	2	1
	PC	Linear Integrated Circuits and Applications	3	0	0	3	3
	PC	Linear Integrated Circuits and Applications Laboratory	0	0	2	2	1
	PC	Digital Communication	3	0	0	3	3
	PC	Digital Communication Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VI	OE	Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Computer and Communication Networks	3	0	0	3	3
	PC	Computer and Communication Networks Laboratory	0	0	2	2	1
	PC	Computer Organization and Microprocessor	3	0	0	3	3
	PC	Computer Organization and Microprocessor Lab	0	0	2	2	1
	PC	Microwave and Antenna	3	0	0	3	3
	PC	Microwave and Antenna Lab	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 4 (3-0-0)	3	0	0	3	3
	PE	Program Elective 5 (3-0-0)	3	0	0	3	3
	PC	Modern Wireless Communication	3	0	0	3	3
	PC	Modern Wireless Communication Lab	0	0	2	2	1
	PC	Embedded System Design	3	0	0	3	3
	PC	Embedded System Design Lab	0	0	2	2	1
	PC	Digital CMOS VLSI Design	3	0	0	3	3
	PC	Digital CMOS VLSI Design Lab	0	0	2	2	1
		Seminar					1
			18	**0**	**6**	**24**	**24**
Sem VIII		**Course Name**					
	Pro	Major/Comprehensive Project					**12**

Program Core Electives:

Semester	VLSI/Embedded systems	Communication Systems	Computer systems	Control
5	Digital Systems Design using HDL	Opto Electronics and Optical Communication		
6	Analog IC Design	Satellite Communication	Machine Learning and Applications	Introduction to Robotics
	Power Electronics	Information Theory and Coding	Database Management Systems	Modern Control Systems
7	Advanced Processors and SoCs	Advanced Communication Networks	Internet of Things	Drones: Design, Theory and Applications
	IC Technology	Radar and Navigation Systems	Deep Learning and Applications	
	Mixed Signal VLSI Design			

Semester 5

<Course Code>					Linear Integrated Circuits and Applications					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To study the basic principles, configurations and practical limitations of op-amp.
- To understand the various linear and non-linear applications of op-amp.
- To understand and design op-amp based active filters.
- To analyze and design op-amp oscillators and waveform.
- to analyze and design circuits based on 555 timer ic and ota.

UNIT I: BASICS OF OPAMP	**8 Hrs.**
Introduction to Op-Amp: Differential amplifier using BJT, Block diagram of op-amp, pin diagram of 741 IC, characteristics of ideal Op-Amp, equivalent circuit of Op-Amp, ideal voltage transfer curve, Op-Amp ac and dc parameters, Virtual ground	
UNIT II: OPAMP APPLICATIONS	**14 Hrs.**
Inverting and non- inverting amplifiers, Voltage follower, Differential amplifier using single and two op-amp, Instrumentation amplifier using Op-Amp, Adder, Voltage to current with floating & grounded load, current to voltage converter, practical integrator & differentiator, Clipping & Clamping circuits, Comparators, log/antilog circuits using Op-Amps, precision rectifiers (half & full wave), Peak detector, Inverting & non inverting Schmitt trigger circuit. Active RC Filters: Idealistic & Realistic response of filters (Low pass, High pass, Band pass and Band reject), Notch Filter.	
UNIT III: WAVEFORM GENERATORS:	**10 Hrs.**
Barkhausen criteria of oscillations, conditions for oscillation, Sine wave generator (Phase shift, Wein bridge, Hartley, Colpitts, and Crystal oscillator), Square and triangular waveform generators (period and frequency analysis), Saw tooth wave generator, Astable, Monostable and Bistable Multivibrators.	
UNIT IV: ADVANCED LICs	**10 Hrs.**
555 Timer IC: Pin-Diagram of 555 Timer IC, Functional block diagram of 555 Timer, Application of 555 Timer as Astable, Bistable and Monostable Multivibrators. Introduction to OTA, OTA as Integrator & Differentiator, IC Analog Multiplier, IC phase locked loops, IC Voltage Regulators, IC VCO, Analog-to-Digital Converter, Digital-to-Analog Converter	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember basic characteristics of opamp and 555 timer.
CO2	:	Understand op-amp properties and limitations.
CO3	:	Apply the knowledge earned to design current mirrors and instrumental amplifiers.
CO4	:	Analyze the circuit performance of op-amp based filter circuits.
CO5	:	Evaluate the design of op-amp‟s linear and non-linear applications.
CO6	:	Create the op-amp based oscillators, filters, waveform generators and ota.

TEXT/REFERENCE BOOKS

1. Ramakant a gayakwad, “op-amps and linear integrated circuits,” pearson, 4th ed, 2015.
2. Salivahanan, bhaaskaran, “linear integrated circuits,” mcgraw hill co. 2nd ed,2017.
3. Robert f. Coughlin, and driscoll,” operational amplifiers and linear integrated circuits”, 6/e, pearson education. Reprint 2007.
4. P. R. Gray and r. G. Meyer,” analysis and design of analog integrated circuit,” john wiley, 4th ed, reprint 2009.

<Course Code>					Linear Integrated Circuits and Applications Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To study the basic principles, configurations and practical limitations of op-amp.
- To understand the various linear and non-linear applications of op-amp.
- To understand and design op-amp based active filters.
- To analyze and design op-amp oscillators and waveform.
- To analyze and design circuits based on 555 timer ic and ota.

LIST OF EXPERIMENTS	
1	Introduction of lab and simulation tool.
2	To measure non idealities in op-amp such as non-infinite input impedance, non-zero output impedance, common-mode rejection ratio etc.
3	To design inverting non inverting op-amp amp amplifier.
4	To design op-amp based rectifiers.
5	To design op-amp comparators.
6	To design op-amp log/antilog amplifier.
7	To design wilson & widlar current mirrors.
8	To design op-amp wave shaping circuits.
9	To design waveform generators: using op-amp.
10	To design 1st and 2nd order lpf and hpf.
11	To design 1st and 2nd order bpf and brf.
12	To design and simulate ic voltage regulators.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember basic characteristics of opamp and 555 timer
CO2	:	Understand op-amp properties and limitations
CO3	:	Apply the knowledge earned to design current mirrors and instrumental amplifiers.
CO4	:	Analyze the circuit performance of op-amp based filter circuits.
CO5	:	Evaluate the design of op-amp‟s linear and non-linear applications.
CO6	:	Create the op-amp based oscillators, filters and waveform generators.

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1. Ramakant a gayakwad, “op-amps and linear integrated circuits,” pearson, 4th ed, 2015.
2. Salivahanan, bhaaskaran, “linear integrated circuits,” mcgraw hill co. 2nd ed,2017.
3. Robert f. Coughlin, and driscoll,” operational amplifiers and linear integrated circuits”, 6/e,pearson education. Reprint 2007
4. P. R. Gray and r. G. Meyer,” analysis and design of analog integrated circuit, john wiley, 4th ed, reprint 2009.

24ICxxxT					Digital Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	0	0	100

COURSE OBJECTIVES

- ➢ To impart basic knowledge of digital modulation and demodulation techniques.
- ➢ To Understand coding and multiple access principles.
- ➢ Understand basic principles of spread spectrum communication system.

UNIT-1: FUNDAMENTAL DIGITAL COMMUNICATION PARAMETERS 12 Hrs.
Sources and Signals, Basic Signal Processing, Sampling Theorem; Quadrature Sampling of Band-Pass Signals. Practical Aspects of Sampling and Signal Recovery, Time-Division Multiplexing, Uncertainty, Information, Entropy, Source Coding Theorem, Huffman Coding, Discrete Memoryless Channels, Mutual Information, Channel Capacity, Channel Coding Theorem.

UNIT-2: OPTIMAL DESIGN OF DIGITAL RECEIVER AND WAVEFORM CODING TECHNIQUES 11 Hrs.
Model of Digital Communication System, Gram-Schmidt Orthogonalization, Geometric Interpretation of Signals, Detection of known Signal in Noise, Correlation Receiver, Matched Filter Receiver, Probability of Errors, Pulse Amplitude Modulation, Pulse Width Modulation, Pulse Position Modulation, Pulse Code Modulation, Differential Pulse-Code Modulation, Delta Modulation, Inter-symbol Interference, Eye Pattern.

UNIT-3: DESIGN OF DIGITAL COMMUNICATION TRANSCEIVER SYSTEM 10 Hrs.
Coherent Binary Modulation Techniques, Quadrature Amplitude Modulation, M-ary Modulation, Effect of Inter-symbol Interference, Bit and Symbol Error Probabilities, Synchronization, Linear Block Codes, Cyclic Codes, Convolution Codes, Maximum-Likelihood Decoding of Convolution Codes.

UNIT 4: ADVANCED CONCEPTS AND WAY FORWARD 9Hrs.
Spread Spectrum Communication, Direct Sequence Spread Spectrum (DSSS), Frequency Hopping Spread Spectrum (FHSS), Introduction to multi-carrier communication systems (OFDM and MIMO), Introduction to Software Defined Radio (SDR).

Max. Hrs.: 42

COURSE OUTCOMES

CO1- Remember principles of various digital communication methods.
CO2 - Understand digital modulation, detection, coding, digital transmission and access techniques.
CO3 - Apply mathematical concepts to model digital communication system.
CO4 - Analyze digital communication systems to obtain various parameters like bandwidth, data rate etc.
CO5 - Evaluate and compare performance of digital communication systems.
CO6 - Design digital communication system at block diagram level.

TEXT/REFERENCE BOOKS

1. Digital Communication - Simon Haykin, 2007, Wiley.
2. Digital Communications-John G Proakis, 2001
3. B. P. Lathi, Modern Digital and Analog Communication Systems, Oxford University Press., 4th edition.

					Digital Communication Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To impart knowledge of digital modulation, demodulation and coding techniques.
- To verify performance of various baseband and carrier digital communication systems.
- To compare various digital communication systems and understand their applications.

LIST OF EXPERIMENTS	
1	PAM, PPM, PWM baseband techniques
2	Sampling, Quantization, PCM and variants
3	Digital carrier communication (ASK, FSK, PSK etc.) techniques
4	TDM techniques and synchronization
5	Generation of digital modulated/demodulated waveforms such as, BPSK, and QPSK using MATLAB and Simulink.
6	System performance in presence of noise. Probability of error calculation and scatterplot.
7	Generation of digital modulated/demodulated QAM waveform and its performance using MATLAB and Simulink.
8	Investigation of the characteristics of Matched Filter and implement Matched Filter based signal detection. Investigations of the eye diagrams.
9	Simulation of baseband and band-pass digital communication system in MATLAB and Simulink.
10	Convolution Codes and Viterbi Detection
11	Spread spectrum communication
12	OFDM Implementation

COURSE OUTCOMES

CO1- Remember principles of various digital communication methods.
CO2 - Understand digital modulation, detection, coding, digital transmission and multiplexing techniques.
CO3 - Apply theoretical and mathematical concepts to practical digital communication system.
CO4 - Analyze and report the characteristics of various modulation, demodulation, multiplexing, synchronization, coding techniques.
CO5 - Evaluate and report performance of digital communication systems.
CO6 - Design and implement small scale digital communication system using development tools.

TEXT/REFERENCE BOOKS

1. B.P.Lathi, Zhi Ding "Modern Digital and Analog Communication Systems", Oxford University Press.
2. P. Chakrabarti, "Analog and Digital Communication", Dhanpat Rai & Co.
3. Wayne Tomasi "Electronic Communications Systems", Pearson education India.
4. Ranjan Bose, "Information Theory, Coding and Cryptography", PHI.
5. T. Rappaport, "Wireless Communications – Principles and Practice",Prentice Hall.

20ECE309T					Control Systems					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To impart fundamental concepts of linear control theory
- To understand the concept of stability of control system and methods of stability analysis in time and frequency domain.
- To facilitate the understanding of design of linear controllers.

UNIT I: INTRODUCTION AND MATHEMATICAL MODELLING OF LINEAR CONTROL SYSTEMS	10 Hrs.
Introduction to Linear and Non-linear control systems, Notion of feedback, Open-loop and closed-loop control systems, Examples of Control Systems, Design and Compensation of Control Systems; Block diagram and signal-flow graphs representations and simplifications; Mathematical modelling and analogy, Differential equations, Transfer Function and Impulse-Response Function, Systems with transport lag, Linearization.	
UNIT II: TIME DOMAIN PERFORMANCE AND STABILITY ANALYSIS	**12 Hrs.**
Time response, Transient response, Steady-state response, Responses of first- and second-order systems to standard test signals and related specifications, Stability definition and meaning, Routh-Hurwitz stability criterion, Analysis of control systems using root locus plots.	
UNIT III: FREQUENCY DOMAIN ANALYSIS AND DESIGN	**12 Hrs.**
Nyquist stability criterion – Gain and Phase margins, Bode plots, Frequency domain specifications and their correlations with time response, Design and performance goals - Steady-state, transient and robustness specifications, Introduction to P, PD, PI and PID controllers and their design, Introduction to Lag and Lead compensator design, Analogy with electronic systems analysis and design.	
UNIT IV: INTRODUCTION TO STATE SPACE ANALYSIS AND WAY FORWARD	**08 Hrs.**
Modelling in State Space, State-Space Representation and Analysis, Canonical state variable model, Equivalence to transfer function model and conversions, Solution of state equations, Concepts of Controllability and Observability, Introduction to modern control systems, and way forward.	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Identify types of control systems.
CO2	:	Understand analysis tools for linear control systems.
CO3	:	Analyze linear control systems using various mathematical tools
CO4	:	Apply concepts in design of linear control systems.
CO5	:	Evaluate performance of linear control systems.
CO6	:	Design linear controllers and compensators.

TEXT/REFERENCE BOOKS

1. I.J. Nagrath and M. Gopal, " Control System Engineering," Wiley Eastern.
2. K. Ogata, “Modern control engineering”, PHI.
3. B C Kuo, “Automatic control systems”, PHI.
4. Norman S. Nise, " Control Systems Engineering," John Wiley.
5. G. F. Franklin, J. D. Powell, and A. E. Emami-Naeini, “Feedback Control of Dynamic Systems”, Pearson.

20ECEXXXP					Control Systems Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

Course Objective (CO)

- To make the students acquainted with the use of computational software such MATLAB, Simulink, and Python to use the control systems analysis and design aspects
- To study the dynamical systems of an interdisciplinary nature and model it in the simulations.
- To impart knowledge on practical aspects of the design of any control system from start to end
- To examine the system performance and modify the system behavior for its states and outputs as desired.
- To impart knowledge of control systems aspects for automation systems

List of Experiments

1. Introduction to MATLAB with respect to Control Systems Applications and Analysis
2. Introduction to Python Programming and Libraries for Control Systems
3. Mathematical modeling of a simple pendulum system using Python and MATLAB
4. Mathematical modeling of a spring-mass damper system using Python and MATLAB
5. Modeling of a slosh-container system in Simulink and Understanding its Mechatronics implementation
6. Transient response analysis of first and second-order control systems
7. Simulating open loop systems using their linear and nonlinear state equations by ODE solvers in MATLAB
8. Design, realize, and simulate an open-loop state-space model for linear and nonlinear systems in Simulink
9. Study the stability analysis using root locus and bode plots using MATLAB
10. Implement Linear Control Systems in MATLAB/Simulink using Libraries and Control System toolbox
11. Design and Implement a PID Controller in Python and Simulink for a linear dynamical system
12. Simulate a simple 2D mobile robot Ground/Aerial in Python and MATLAB/Simulink in an open loop
13. Design and implement a simple State-feedback controller for a robot
14. Implementation of Simulink-based Kinematic model of an autonomous vehicle.
15. Implementation of a Control algorithm for an Industrial Automation System
16. Build an autonomous system with Control principles

Course Outcome (CO)

After the completion of the course, students will be able to,

CO1. Remember the basic concepts of control systems.
CO2. Understand the use of various programming paradigms for control systems different types of coding
CO3. Apply various control system specific tools and functions to model the dynamical system
CO4. Analyze the control system behaviors and its performance through simulations
CO5. Evaluate and compare various specified system responses and its stabilities
CO6. Develop, implement and build various controllers to perform the closed-loop analysis for practical control systems and robots.

Core Electives

Semester 5

<20ECE305T >					Optoelectronics and Optical Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To introduce the students to various optical fiber modes, configurations, and various signal degradation factors associated with optical fiber
- To understand various optical sources and optical detectors
- To understand the optical communication system, optical amplifiers.
- To study different fiber network elements and basic optical components.

UNIT I: BASICS OF OPTICAL FIBER	**10 Hrs.**
Introduction to Optical Fiber and Waveguides, TEM, TE and TM mode in wave guide, Wave Propagation in Step-Index & Graded-Index Fiber, Modes & Rays. Basic Optical Communication System, Advantage and applications of Optical Communication System.	
UNIT II: LOSSES IN OPTICAL FIBERS	**10 Hrs.**
Introduction, Absorption, Scattering, Overall Fiber Dispersion-Single Mode Fibers, Multimode Fibers, Dispersion-Shifted Fiber, Dispersion-Flattened Fiber, Polarization.	
UNIT III: SOURCES & DETECTORS	**10 Hrs.**
Couplers, Isolators and Circulators, Multiplexers and Filters, Lasers, Light-Emitting Diodes, Photodetectors, Switches, Wavelength Converters.	
UNIT IV: OPTICAL NETWORKS	**12 Hrs.**
SONET/SDH, Optical Transport Network, Optical Access Network, Ethernet, Network Topologies. Advanced Multiplexing Strategies- OFDM, SDM. Subcarrier Multiplexing, WDM Network Architectures, Few Advanced Topics: Flexible Optical Networks, Optical Neural Network, Cognitive Optical Network.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember the basic concepts of optical fiber structure and optical communication systems.
CO2	:	Understand different design aspects of optical fiber including its advantages and limitations.
CO3	:	Apply optical fiber and optical communication concepts.
CO4	:	Analyze the system performance of optical transmitters, receivers, and optical amplifiers.
CO5	:	Evaluate the design of optical fiber communication systems.
CO6	:	Create optical communication applications in the recent technological trends.

TEXT/REFERENCE BOOKS

1. Gerd Keiser, “Optical Fiber Communications”, 4th Edition McGraw Hill.
2. John M. Senior, “Optical Fiber Communication” PHI/Pearson.
3. G. P. Agrawal, “Fiber optic Communication Systems”, John Wiley and sons.
4. John Gowar, “Optical Communication Systems” Pearson.

24XXXXXX					Digital Systems Design Using HDL					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	0	0	100

COURSE OBJECTIVES

- To Learn and understand the basics of Hardware description language its use in designing digital circuits.
- To get an in-depth knowledge Digital system design options and trade-offs.

UNIT 1: REVIEW OF COMBINATIONAL AND SEQUENTIAL DESIGN **10 Hrs**

Review of Combinational and Sequential logic design, Moore and Mealy machine, Design of synchronous sequential circuits – state diagram, state table, state table assignment and reduction, design of iterative circuits – ASM chart and realization of ASM chart, Timing issues in sequential circuits.

UNIT 2: INTRODUCTION TO VERILOG HDL AND MODELLING STYLE: GATE LEVEL AND DATA FLOW **10 Hrs**

Introduction, Language Elements, Expressions, Modules and Ports, Built-in Primitives, User Defined Primitives, Different Modelling Styles, Tasks and Functions, Test bench., **Gate level modelling:** Introduction, AND Gate Primitive, Module Structure, Other Gate Primitives, Illustrative Examples, Tri-State Gates, Array of Instances of Primitives, Additional Examples, Design of Flipflops with Gate Primitives, Delays, Strengths and Contention Resolution, Net Types, Design of Basic Circuits, **Dataflow modelling:** Introduction, Continuous Assignment Structures, Delays and Continuous Assignments, Assignment to Vectors, Operators.

UNIT 3: MODELLING STYLE: BEHAVIOURAL MODELLING **11 Hrs**

Introduction, Operations and Assignments, Functional Bifurcation, Procedural Assignments: Initial and Always Construct, Examples, Assignments with Delays, wait construct, Multiple Always Blocks, Designs at Behavioural Level, Blocking and Non-blocking Assignments, The case statement, Simulation Flow. if and if-else constructs, assign-deassign construct, repeat construct, for loop, the disable construct, while loop, forever loop, parallel blocks, force-release construct, Event, Memory modelling: RAM, ROM.

UNIT 4: DESIGNING WITH FPGAS **11 Hrs**

Introduction to FPGA architectures: Overview, programming technologies, Logic block architecture: FPGA logic cells, timing models, power dissipation, I/O block architecture: Input and Output cell characteristics, clock input, Timing, Power dissipation, Programmable interconnect - Partitioning and Placement, Routing resources, delays, RTL Coding Styles for Synthesis, Analysing the RTL Design, Timing Closure Challenges, Design Sign-off.

Max. Hrs.: 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Remember the concept of digital logic design.
CO2 - Understand the basics of Verilog HDL, program structure and basic language elements.
CO3 – Explain various modelling styles to Verilog HDL.
CO4 – Analyse the efficient modelling style to model various combinational and sequential circuits.
CO5 - Evaluate the performance parameters for a digital circuit.
CO6 – Design efficient digital circuits on a FPGA platform.

TEXT/REFERENCE BOOKS

1. Samir Palnitkar, “Verilog HDL A Guide to Digital Design and Synthesis”, Pearson Education.
2. Michael D. Ciletti, "Advanced Digital Design with the VERILOG HDL”, Pearson Education.
3. Stephen Brown, Zvonko Vranesic, "Fundamentals of Digital Logic with Verilog", McGraw-Hill-2007.
4. Richard C. Dorf, John V. Oldfield “Field-Programmable Gate Arrays: Reconfigurable Logic for Rapid Prototyping and Implementation of Digital Systems”. Wiley, 2008